S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 03

Aim: Automate student marksheet generation, system information display, Fibonacci and prime number generation, and file management operations using shell scripts to enhance computational efficiency and user interaction.

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✞ **Aim:** Automate student marksheet generation, system information display, Fibonacci and prime number generation, and file management operations using shell scripts to enhance computational efficiency and user interaction.

✞ **Tasks to be done in this Practical.**

a) Write a shell script to generate mark- sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student.

b) Write a menu driven shell script which will print the following menu and execute the given task.
   - Display calendar of current month.
   - Display today's date and time.
   - Display usernames those are currently logged in the system.
   - Display your terminal number

c) Write a shell script which will generate first n Fibonacci numbers like: 1, 1, 2, 3, 5, 13

d) Write a shell script which will accept a number b and display first n prime numbers as output.

e) Write menu driven program for file handling activity ○ Creation of file.
   - Write content in the file. ○ Upend file content.
   - Delete file content

✞ **Objectives:**

1. Automate marksheet generation with total, percentage, and class classification.
2. Develop menu-driven scripts for system information and file operations.
3. Generate Fibonacci and prime numbers for user-defined inputs.

✞ **Requirements:**

Hardware Requirements:

- Processor: Minimum 1 GHz
- RAM: 512 MB or higher
- Storage: 100 MB free space

Software Requirements:

- Operating System: Linux/Unix-based
- Shell: Bash 4.0 or higher
- Text Editor: Nano, Vim, or any preferred editor

✞ Theory:

Shell scripting is a powerful way to automate repetitive tasks and manage system operations efficiently. It allows users to write programs using shell commands and scripting constructs. Shell scripts are interpreted line-by-line by a shell interpreter, making them ideal for administrative tasks, file management, and system automation. This practical encompasses a variety of real-world scenarios that demonstrate the utility of shell scripting for computing tasks and resource management.

1. Marksheet Generation

This script takes input marks for three subjects, calculates the total marks, percentage, and determines the class of the student based on predefined conditions. Conditional statements (if-else) are used to classify the performance into distinction, first class, second class, or fail. This exercise emphasizes the use of arithmetic operations and decision-making constructs.

Key concepts include:

- Reading user input using read
- Arithmetic operations with $((expression))
- Conditional statements for decision-making

2. Menu-Driven Script for System Information

Menu-driven scripts enhance user interaction by presenting a list of options for performing different tasks. In this practical, options are provided to display the calendar of the current month, the current date and time, logged-in users, and the terminal number. The script utilizes looping constructs (while) and case statements for structured flow control.

Commands used:

- cal for displaying the calendar
- date for showing current date and time
- who to list logged-in users
- tty to identify the terminal

3. Fibonacci Number Generation

Fibonacci numbers are a sequence where each term is the sum of the two preceding ones. The script uses iterative constructs (for loop) to generate n terms based on user input. This practical illustrates the use of loop control and variable swapping to generate series data efficiently.

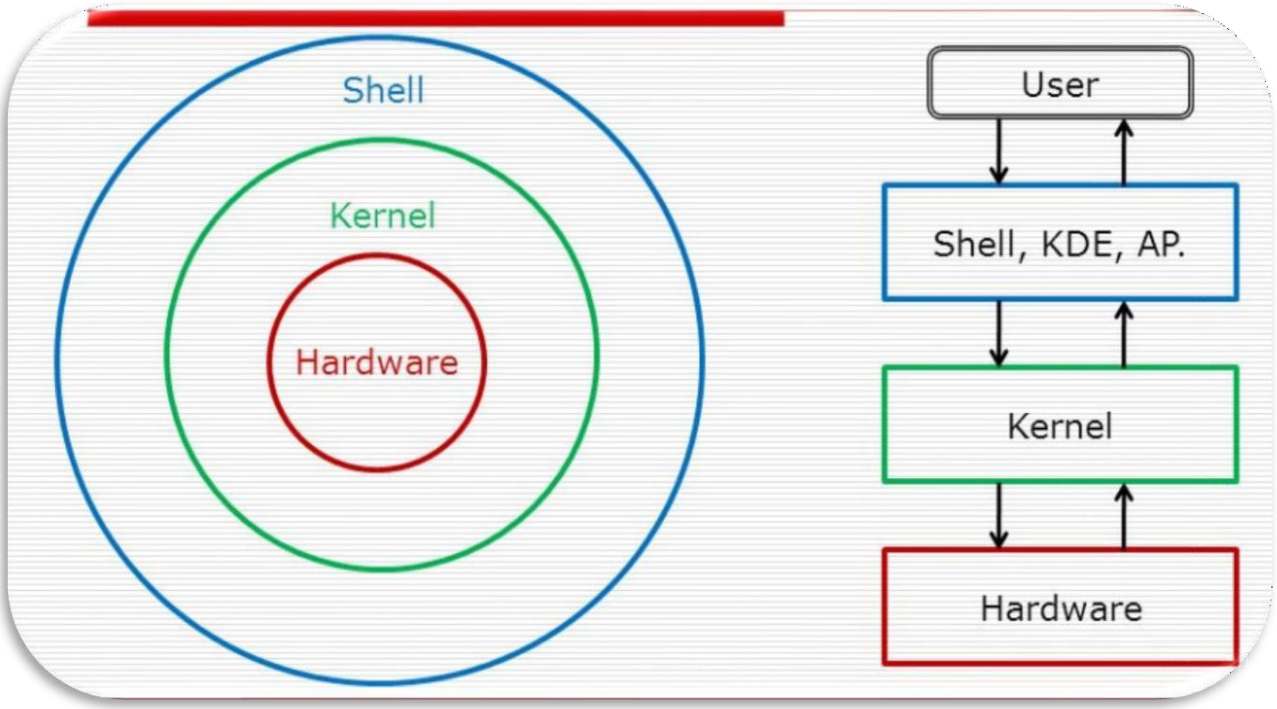
4. Prime Number Display
This script accepts an integer n and outputs the first n prime numbers. A nested loop checks divisibility to determine if a number is prime. The practical demonstrates logic building for number-theoretic operations using loops and conditionals.

5. Menu-Driven File Management
The file handling script enables users to create, write, append, and delete file content. The case construct manages different file operations.
Commands include:

- touch to create files
- cat for writing and appending content
- rm for deleting files

This exercise emphasizes text manipulation, input handling, and file control mechanisms in Unix-like environments.

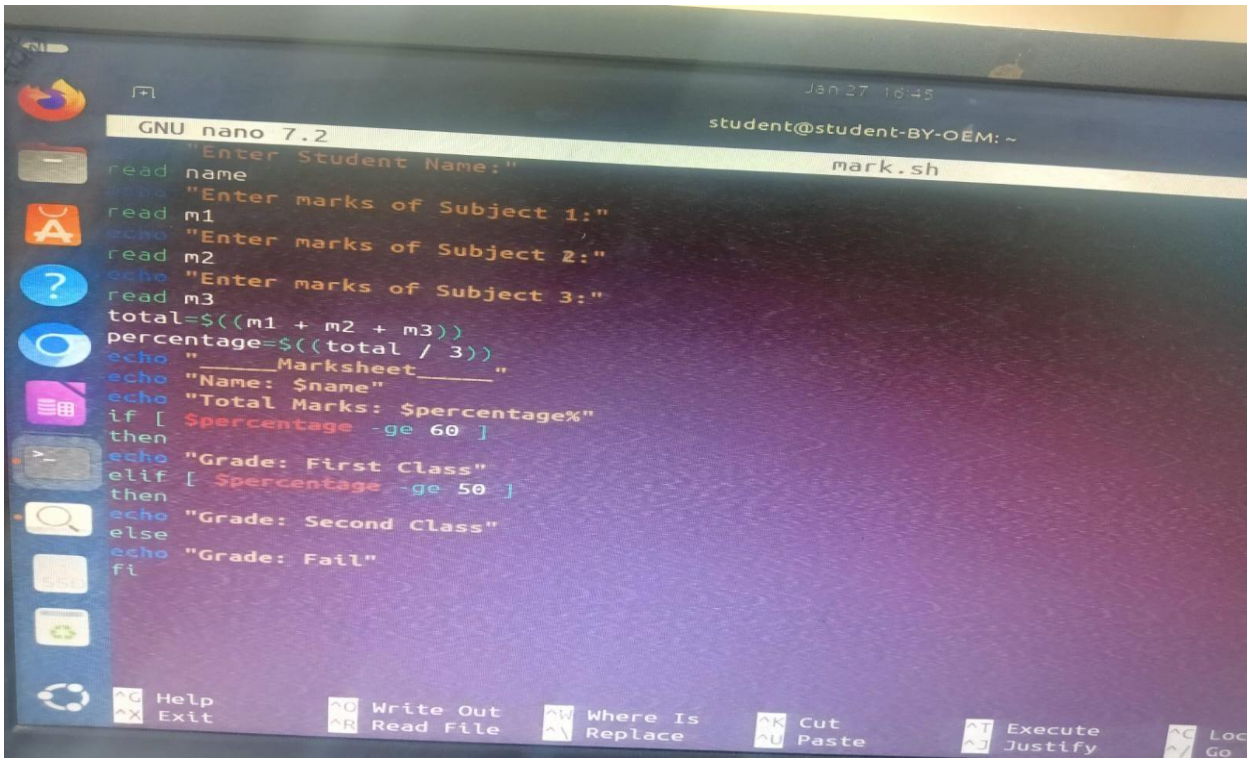
Diagrammatical View of Shell

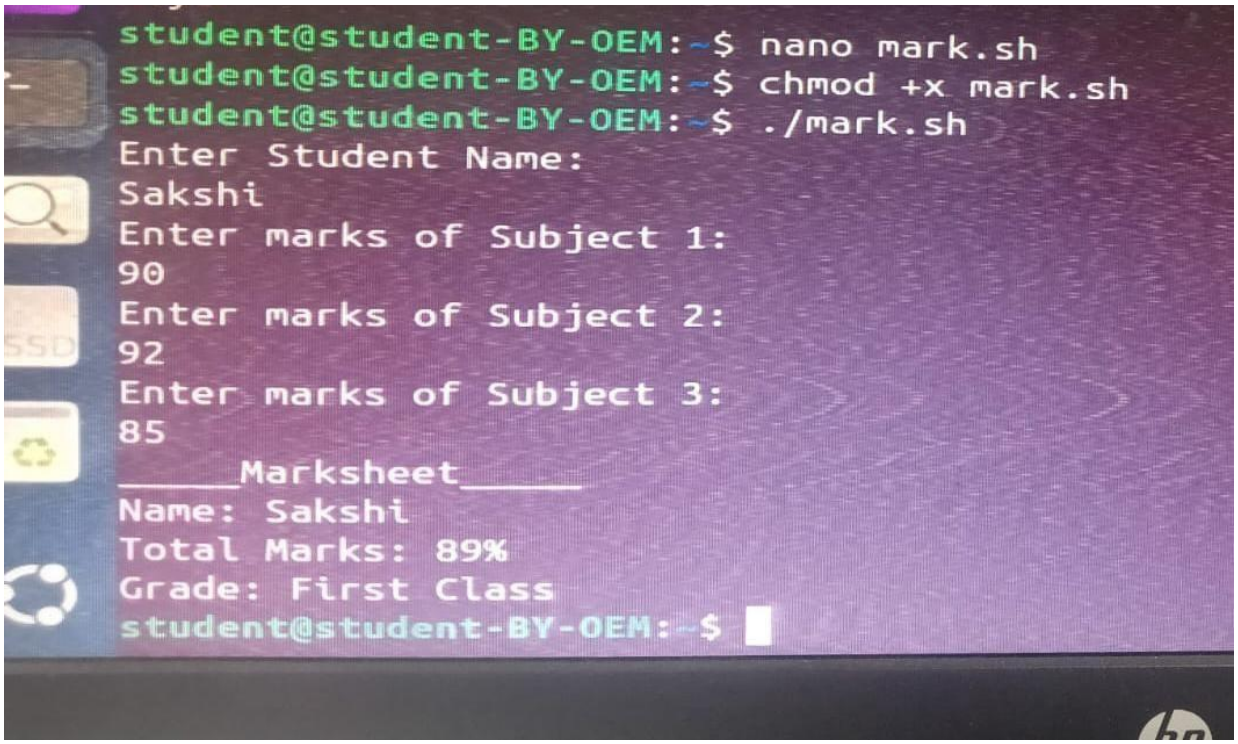


✞ CODES

1. **Write a shell script to generate mark- sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student.**

Output 1:





2. **Write a menu driven shell script which will print the following menu and execute the given task.**
   - **Display calendar of current month.**
   - **Display today's date and time.**
   - **Display usernames those are currently logged in the system.**
   -
   - **1Display your terminal number**

Output 2:

```
echo "_____ System Information _____"
echo "Username: $(whoami)"
echo "Hostname: $(hostname)"
echo "Current Date: $(date)"
echo "Uptime:"
uptime
echo "Disk Usage:"
dff -h
```

```
VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~
$ git --version
git version 2.41.0.windows.1

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~
$ mkdir shell-practical
mkdir: cannot create directory 'shell-practical': File exists

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~
$ cd shell-practical

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ git init
Reinitialized existing Git repository in C:/Users/VAISHNAVI/shell-practical/.git/

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ nano Sakshi.txt

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ chmod +x Sakshi.sh
chmod: cannot access 'Sakshi.sh': No such file or directory

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ chmod +x Sakshi.txt

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ ./Sakshi.sh
bash: ./Sakshi.sh: No such file or directory

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ ./Sakshi.txt
_____ System Information _____
Username: VAISHNAVI
Hostname: LAPTOP-G4BBT1LV
Current Date: Tue Jan 27 21:32:23 IST 2026
Uptime:
./Sakshi.txt: line 6: uptime: command not found
Disk Usage:
./Sakshi.txt: line 8: dff: command not found

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$
```

3. **Write a shell script which will generate first n Fibonacci numbers like: 1, 1, 2, 3, 5, 13**

Output 3:

```
echo "Enter number of terms:"
read n
a=0
b=1
echo "Fibonacci Series:"
for (( i=0; i<n; i++))
do
echo -n "$a "
fn=$((a+b))
a=$b
b=$fn
done
```

```
VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ chmod +x fibonacci.txt

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ ./fibonacci.txt
Enter number of terms:
7
Fibonacci Series:
0 1 1 2 3 5 8
```

4. Write a shell script which will accept a number b and display first n prime numbers as output.

Output 4:

```
  GNU nano 7.2
echo "Enter a number:"
read num
flag=0
for (( i=2; i<=num/2; i++ ))
do
if [ $((num % i)) -eq 0 ]
then
flag=1
break
fi
done
if [ $flag -eq 0 ]
then
echo "$num is a prime number"
else
echo "$num is not a prime number"
fi
```

```
VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ nano prime.txt

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ chmod +x prime.txt

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ ./prime.txt
Enter a number:
7
7 is a prime number
```

5. Write menu driven program for file handling activity
   - Creation of file.
   - Write content in the file.
   - Upend file content.
   - Delete file content

Output 5:

```
  GNU nano 7.2
echo "1. Create File"
echo "2. Delete File"
echo "3. Rename File"
read ch
case $ch in
1)
echo "Enter file name:"
read fname
touch $fname
echo "File created"
;;
2)
echo "Enter file name:"
read fname
touch $fname
echo "File deleted"
;;
3)
echo "Enter old file name:"
read old
echo "Enter new file name:"
read new
mv $old $new
echo "File renamed"
;;
*)
echo "Invalid choice"
;;
esac
```

```
VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ chmod +x file.txt

VAISHNAVI@LAPTOP-G4BBT1LV MINGW64 ~/shell-practical (master)
$ ./file.txt
1. Create File
2. Delete File
3. Rename File
1
Enter file name:
file.txt
File created
```

✞ **Conclusion**: Shell scripting helps automate tasks like marksheet generation, system info display, number generation, and file management, making operations faster, efficient, and user-friendly.

✞ **Discussion Questions:**

1. **What is the purpose of using shell scripting in this practical?**
2. **Which command is used to display the current date and time?**
3. **How does the script calculate the Fibonacci sequence?**
4. **Which command is used to create a file in the file management script?**
5. **How does the prime number script determine if a number is prime?**

✞ **References:**

https://www.tutorialspoint.com/unix/shell_scripting.html
https://www.javatpoint.com/shell-scripting-tutorial

Date:____/____/2026

Signature
Course Coordinator